

TacBPM: A Tactile-conditioned Behavior Prior Model for Dexterous Reorientation

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Abstract—Dexterous in-hand manipulation requires policies that coordinate many coupled joints through intermittent, contact-rich interaction. Although reinforcement learning can train strong specialist policies, each new object family, target distribution, or sensing configuration often requires costly exploration in a high-dimensional hand-action space. We propose TacBPM, a tactile-conditioned behavior prior model for dexterous reorientation. TacBPM first trains eight sphere reorientation specialists over scale factors from 0.3 to 1.0 relative to a nominal 8 cm-diameter sphere, and then distills their closed-loop behavior into a variational latent controller with a tactile-conditioned prior and decoder. The prior conditions on proprioceptive and tactile history, including per-finger contact magnitudes, contact locations, and object scale cues, so latent codes remain grounded in the current hand-object interaction state. Downstream policies keep the prior fixed and act through residual latent actions; the decoder is either frozen for strict reuse or finetuned for new geometry. Across held-out scales, anisotropic objects, commanded-axis rotation, and real-robot rollouts, this prior-centered interface improves transfer while preserving contact-stable behavior.

Index Terms—Dexterous manipulation, behavior priors, tactile sensing, policy distillation, reinforcement learning.

I. INTRODUCTION

DEXTEROUS robotic hands can rotate, stabilize, and regrasp objects without external fixtures, making them relevant to service robots, industrial manipulation, assistive systems, and teleoperation. Real deployments, however, require control policies to handle variation in shape, scale, pose, friction, and sensing quality. This setting remains difficult because successful in-hand manipulation is not only a sequence of joint targets; it is a closed-loop contact strategy that must adapt to object motion, hand configuration, and fingertip feedback.

Large-scale reinforcement learning has produced capable dexterous manipulation policies in simulation [1], [2]. With appropriate rewards, domain randomization, and privileged object information, specialist policies can solve challenging reorientation tasks. The limitation is reusability: a policy trained for one object family, target distribution, or observation setting typically transfers poorly as an action interface for another task. Downstream learning must therefore re-explore from low-level joint commands, where random actions often break contact, drop the object, or enter unstable hand configurations.

A reusable behavior prior can reduce this exploration burden by exposing a compact latent action to downstream policies. Prior work in physics-based control shows that motor skills can be distilled into encoder-decoder priors, allowing downstream reinforcement learning to explore through residual latent codes rather than raw joint targets [3]. Dexterous manipulation requires the same structure, but the prior must be contact-aware. An effective hand prior should change when a fingertip

gains or loses contact, force shifts across fingers, or the object begins to slip.

We propose TacBPM, a tactile-conditioned behavior prior model for dexterous in-hand reorientation. TacBPM first trains eight sphere specialists with scale factors from 0.3 to 1.0 relative to a nominal 8 cm-diameter sphere. This controlled teacher family exposes the hand to different apertures, contact-force patterns, and rolling/regrasping regimes while keeping geometry simple. These specialists are then distilled into a variational latent controller with an encoder $q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{g})$, a learned prior $p_\theta(\mathbf{z}|\mathbf{x})$, and a decoder $\pi_\psi(\mathbf{a}|\mathbf{x}, \mathbf{z})$. Unlike a purely proprioceptive prior, TacBPM conditions $\mathbf{x}$ on hand state, previous targets, tactile contact magnitudes, tactile contact positions, and object scale cues over a short history.

Downstream policies keep the tactile prior fixed and output a residual latent action $\Delta\mathbf{z}$ from task observations. The executed latent is $\mathbf{z} = \mu_{\text{prior}} + \Delta\mathbf{z}$, which the decoder maps to a 22-dimensional hand command. The decoder is either frozen to evaluate strict reuse or finetuned while the prior remains fixed to adapt latent commands to new object geometries. This design separates reusable contact behavior from task-specific intent: the prior anchors exploration near stable hand-object interactions, and the downstream policy selects low-dimensional corrections.

Our contributions are:

- We formulate tactile-conditioned behavior prior learning as multi-teacher distillation from sphere reorientation specialists.
- We design a tactile-conditioned prior and decoder that adapt latent behavior to contact magnitude, contact position, and object scale.
- We reuse the learned prior for downstream reinforcement learning through residual latent control, with either a frozen decoder or a task-finetuned decoder.
- We evaluate the prior on held-out scales, anisotropic objects, commanded-axis rotation, and real-robot transfer.

II. RELATED WORK

A. Reusable Skill Representations and Dexterous Behavior Priors

Reusable latent action spaces provide structured exploration for high-dimensional control. In physics-based character control, adversarial skill embeddings such as ASE and CALM map random or user-conditioned latents to realistic behavior [4], [5], while NPMP, ControlVAE, PhysicsVAE, and neural categorical priors learn latent motor representations through reconstruction, dynamics modeling, or imitation objectives [6]–[9]. Recent humanoid behavior priors further distill broad motor skills into variational latent spaces with learned proprioceptive priors [3]. Dexterous manipulation

requires similar reusable structure, but object rotation and reorientation policies remain sensitive to object distribution and task objective [10]–[12]. TacBPM addresses this gap by learning a tactile-conditioned behavior prior that remains fixed while downstream policies act through residual latent codes.

B. Tactile and Visuotactile Dexterous Manipulation

Tactile sensing provides direct evidence of contact state, load distribution, and incipient slip. Touch-only and visuotactile manipulation systems use tactile signals for object rotation, reorientation, and translation under occlusion [13]–[16]. Other systems improve transfer through sim-to-real training, rapid motor adaptation, visual perception transfer, touch-aware rotation policies, and learned dynamics adaptation [14], [17]–[22]. Tactile representation learning further shows that contact-rich pretraining, neural visuotactile fields, and unified 3D visuotactile features improve fine-grained manipulation and state inference [23]–[25]. TacBPM differs from kinematics-only priors by conditioning latent behavior on smoothed contact magnitudes and 3D contact positions over time, allowing the latent distribution to change with the current contact pattern.

C. Policy Distillation and Behavior Prior Models

Policy distillation provides a mechanism for transferring behavior from specialized teachers to a reusable student. It has been used for policy compression, policy transfer, and teacher combination [26], while dataset aggregation and kick-starting methods query teachers on the learner-induced state distribution [27]. In dexterous manipulation, curricula, simple-to-complex transfer, and mixture-of-experts designs extend policies beyond a single training regime [28], [29]. TacBPM adopts online teacher labeling, but the transferred object is not a complete downstream actor. Instead, TacBPM distills a state-conditioned tactile prior that remains fixed during downstream learning. The high-level policy predicts residual latent intent, and the decoder either tests strict reuse or adapts the latent-to-action map for new geometry. This separates TacBPM from actor transfer, action-space residual learning, and mixture-of-experts controllers, which primarily reuse or combine executable policies rather than reuse a contact-conditioned action prior.

III. PRELIMINARIES

A. Dexterous Reorientation

We consider a dexterous robotic hand manipulating a rigid object in simulation. At each control step, the environment state includes the hand joint positions, previous control targets, tactile contacts, object pose, and a target object orientation. The policy outputs a 22-dimensional hand action $\mathbf{a} \in \mathbb{R}^{22}$, which is converted to low-level position targets and applied through the simulator. The task reward encourages the object orientation to match the target while penalizing unstable hand motion, excessive work, and object drops.

B. Behavior Prior Learning

Behavior prior learning replaces raw action exploration with structured latent control. Instead of learning a downstream policy $\pi(\mathbf{a}|\mathbf{o}, \mathbf{g})$ directly in the hand-action space, we learn a decoder $\pi_\psi(\mathbf{a}|\mathbf{x}, \mathbf{z})$ and a prior $p_\theta(\mathbf{z}|\mathbf{x})$. Downstream policies output latent actions, and the decoder maps them to hand commands. The decoder is either frozen to evaluate strict reuse or finetuned with the high-level policy while the prior remains fixed. Because downstream rollouts require stable closed-loop behavior, we distill from trained specialists rather than fitting offline action snapshots alone.

IV. METHOD

A. Overview

TacBPM learns a tactile-conditioned behavior prior in three stages. First, sphere reorientation specialists are trained over scale factors from 0.3 to 1.0 of a nominal 8 cm sphere. Second, their closed-loop behavior is distilled into a variational latent controller with an encoder, a learned prior, and a decoder. Third, downstream policies freeze the prior and output residual latent actions for harder objects and tasks; the decoder is either frozen for strict reuse or finetuned for geometry adaptation. This pipeline preserves contact-aware behavior while allowing downstream tasks to express intent through compact latent corrections.

B. Specialist Policy Training

The specialist stage provides controlled contact-behavior coverage. We train sphere reorientation specialists with proximal policy optimization [1] in Isaac Gym/Isaac Lab-style massively parallel simulation [2]. The nominal sphere diameter is 8 cm, and the eight specialists use scale factors $\{0.3, 0.4, \dots, 1.0\}$. Each specialist observes a teacher-compatible state $\mathbf{o}_t$ containing tactile-proprioceptive history, task goal information, and teacher-only privileged object information. Varying scale changes hand aperture, contact distribution, and rolling/regrasping regime while keeping object geometry simple.

Online multi-teacher distillation aligns supervision with the student’s rollout distribution. Environments are assigned to the eight sphere-scale teachers in a strided batch layout. At each step, the assigned teacher produces the corresponding low-level target action, and all targets are interleaved into a single training batch. This DAgger-style procedure preserves scale-specific expert behavior while avoiding an offline action dataset that may not match the student’s induced states.

C. Tactile-Conditioned State Representation

The student state $\mathbf{x}_t$ is a compact tactile-proprioceptive history. Each frame contains normalized hand joint positions, previous control targets, five smoothed tactile contact magnitudes, and five 3D tactile contact positions. We stack the most recent three frames, yielding a 192-dimensional state:

$$\mathbf{x}_t = \left[\mathbf{q}_{t-2:t}, \mathbf{u}_{t-2:t}, \mathbf{c}_{t-2:t}^f, \mathbf{c}_{t-2:t}^p \right], \quad (1)$$

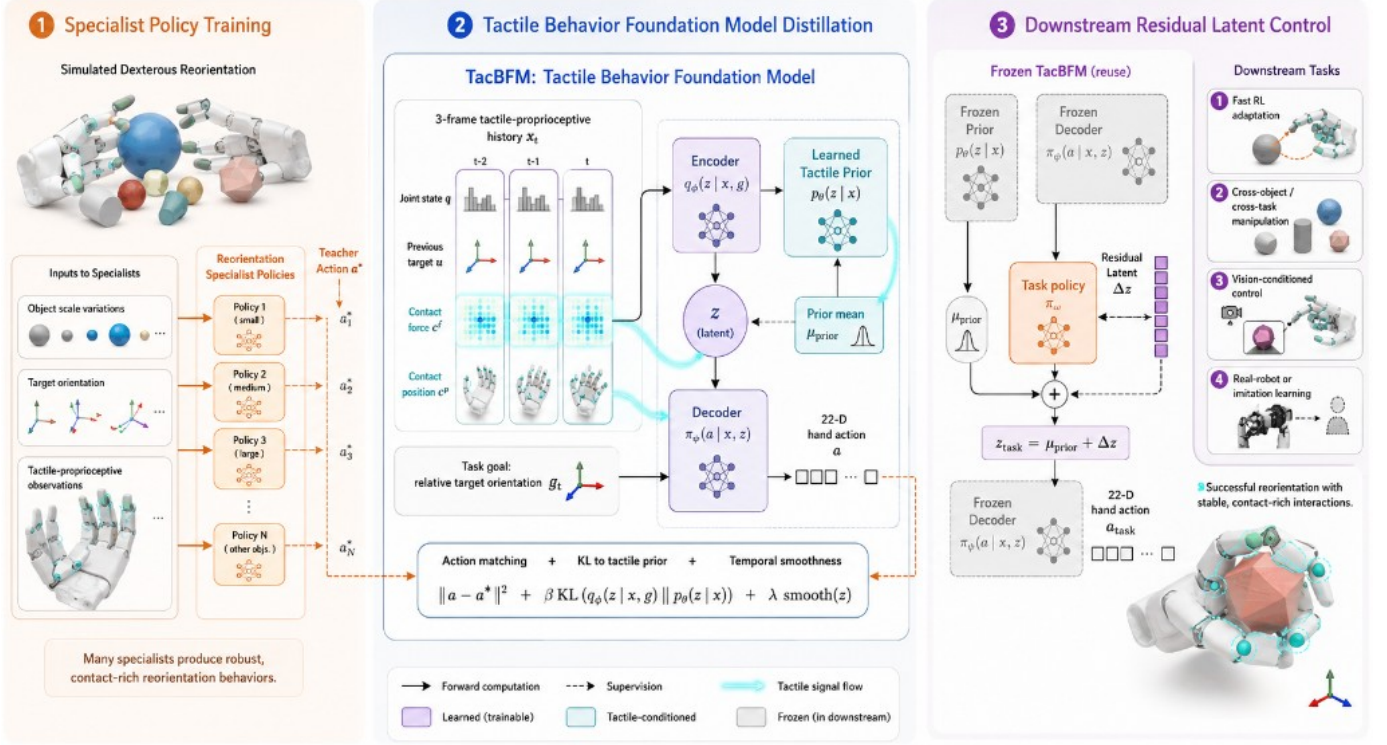


Fig. 1. Overview of TacBPM. The pipeline first trains sphere reorientation specialist policies over multiple scales, distills their tactile-proprioceptive closed-loop behavior into a generalist latent controller with a learned tactile prior, and finally reuses the prior-centered residual latent interface for downstream control on more complex objects and manipulation tasks. The decoder can remain frozen for strict reuse or be finetuned for downstream object geometry.

where $\mathbf{q}$ denotes hand joint state, $\mathbf{u}$ denotes previous control targets, $\mathbf{c}^f \in \mathbb{R}^5$ denotes per-finger contact magnitude, and $\mathbf{c}^p \in \mathbb{R}^{15}$ denotes per-finger 3D contact locations. The task observation $\mathbf{g}_t$ is the relative target orientation in the no-position downstream setting.

The encoder, prior, and decoder process this history with a direct-concatenation MLP, together with an object longest-side scale cue. This design makes $p_\theta(\mathbf{z}|\mathbf{x}_t)$ depend on the current tactile contact pattern and object scale rather than on joint pose alone. Consequently, the same object-orientation goal can induce different latent distributions under stable pinch, weak contact, or impending contact loss.

D. Specialist-to-Generalist Variational Distillation

TacBPM distills specialist actions into an encoder-prior-decoder latent controller. The encoder models the posterior distribution over latent actions conditioned on tactile-proprioceptive state and task goal:

$$q_\phi(\mathbf{z}_t|\mathbf{x}_t, \mathbf{g}_t) = \mathcal{N}(\boldsymbol{\mu}_t^{\text{enc}}, \boldsymbol{\sigma}_t^{\text{enc}}). \quad (2)$$

The learned prior predicts a state-conditioned latent distribution:

$$p_\theta(\mathbf{z}_t|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_t^{\text{prior}}, \boldsymbol{\sigma}_t^{\text{prior}}). \quad (3)$$

The decoder maps the tactile-proprioceptive feature and sampled latent code to a hand action:

$$\mathbf{a}_t = \pi_\psi(\mathbf{x}_t, \mathbf{z}_t). \quad (4)$$

The encoder, prior, and decoder operate on the concatenated tactile-proprioceptive MLP feature and use a 32-dimensional latent space.

During online distillation, the student rolls out in the environment and the assigned sphere-scale specialist provides the teacher action $\mathbf{a}_t^*$ at the same state. The training objective combines action matching, KL regularization to the learned prior, and temporal smoothness over consecutive posterior means:

$$\mathcal{L} = \|\pi_\psi(\mathbf{x}_t, \mathbf{z}_t) - \mathbf{a}_t^*\|_2^2 + \beta D_{\text{KL}}(q_\phi(\mathbf{z}_t|\mathbf{x}_t, \mathbf{g}_t) \| p_\theta(\mathbf{z}_t|\mathbf{x}_t)) + \lambda \|\boldsymbol{\mu}_t^{\text{enc}} - \rho \boldsymbol{\mu}_{t-1}^{\text{enc}}\|_2. \quad (5)$$

The KL term trains the prior to predict task-relevant latent codes without the task goal, and the temporal term discourages discontinuous latent jumps. In the reported configuration, β is annealed from 0.01 to 0.001, $\lambda = 0.005$, $\rho = 0.99$, and $\mathbf{z} \in \mathbb{R}^{32}$.

E. Residual Latent Downstream Control

Downstream learning uses the distilled prior as a fixed action-space anchor. The tactile state normalizer and prior network are frozen. A downstream policy receives the normalized tactile-proprioceptive state and task observation, and predicts a residual latent action $\Delta \mathbf{z}_t$. The executed latent code is

$$\mathbf{z}_t^{\text{task}} = \boldsymbol{\mu}_t^{\text{prior}} + \Delta \mathbf{z}_t, \quad (6)$$

and the environment action is

$$\mathbf{a}_t^{\text{task}} = \pi_\psi(\mathbf{x}_t, \mathbf{z}_t^{\text{task}}). \quad (7)$$

TABLE I

CORE IMPLEMENTATION PARAMETERS. DOWNSTREAM AND AXIS-CONDITIONED POLICIES LOAD THE PRETRAINED GENERALIST CHECKPOINT, FREEZE THE STATE NORMALIZER AND PRIOR NETWORK, AND TRAIN A RESIDUAL-LATENT PPO POLICY. DECODER-FINETUNING VARIANTS UPDATE ONLY THE PRETRAINED DECODER MLP AND OUTPUT HEAD IN ADDITION TO THE HIGH-LEVEL POLICY; THE PRIOR REMAINS FIXED.

Component	Setting
Hand/action interface	Sharpa Wave-style 22-DoF hand; 22-D position-target action
Observation history	3 frames $\times$ 64 values = 192-D tactile-proprioceptive state
Per-frame state	joint state 22, previous targets 22, contact magnitudes 5, contact positions 15
Specialist teachers	8 sphere scales, 0.3–1.0 of a nominal 8 cm sphere
Distillation environments	8192 parallel environments, strided across sphere-scale teachers
Encoder/prior/decoder MLPs	[512, 256, 128] hidden units; Gaussian 32-D latent heads
Distillation optimizer	Adam, learning rate 3×10^{-4} , gradient clip 1.0
Latent regularization	KL weight 0.01 $\rightarrow$ 0.001; temporal loss $\lambda = 0.005$, $\rho = 0.99$
Downstream PPO	8192 environments, horizon 8, minibatch 32768, 5 mini-epochs
PPO hyperparameters	Adam, learning rate 3×10^{-4} , $\gamma = 0.99$, GAE $\tau = 0.95$, clip 0.2
Residual latent policy	policy MLP [512, 256, 128], 32-D residual, residual scale 1.0
Decoder adaptation	frozen for strict reuse; decoder MLP/head trainable for geometry adaptation
Axis command input	7 commands ($\pm x, \pm y, \pm z$, still), one-hot repeated 5 times
Axis training schedule	command resampling every 4 s; still angular-velocity clip 0.5

This residual formulation constrains exploration near the tactile-conditioned prior while still allowing task-driven deviations. A frozen decoder measures strict behavior-prior reuse, whereas decoder finetuning adapts the realization of latent commands to new object geometry. In both cases, the high-level policy uses the same prior-centered latent interface.

F. Implementation Details

Table I summarizes the reported configuration. Each of the last three frames contains 22 normalized joint positions, 22 previous target positions, 5 tactile magnitudes, and 15 contact-position coordinates. The no-tactile ablation uses the same architecture and training objective, but zeros the tactile force and contact-position channels before normalization.

V. EXPERIMENTS

The experiments evaluate whether contact behavior distilled from sphere specialists forms a reusable prior rather than a sphere-specific controller. We organize the evaluation around five questions:

- **Q1. Prior coverage and extrapolation:** Does a single tactile prior preserve the sphere-scale teacher family while transferring to held-out scales and shapes?
- **Q2. Downstream transfer:** With the tactile prior fixed, does residual latent control improve reorientation on anisotropic objects, and does decoder finetuning help adapt to new geometry?
- **Q3. Commanded rotation:** Can the same latent motor substrate support positive and negative rotation about commanded Cartesian axes?
- **Q4. Embodiment transfer:** Does the hand prior remain effective when the hand is controlled as part of a larger arm-hand manipulation system?
- **Q5. Learning interface:** Can TacBPM serve as an action interface for single-environment online reinforcement learning and imitation learning?

A. Experimental Setup

Simulation and task setting. All simulation comparisons use the same dexterous hand model, object initialization protocol, and termination rules within each benchmark. The prior is trained from eight sphere specialists with scale factors $\{0.3, 0.4, \dots, 1.0\}$ of the nominal 8 cm sphere. During distillation, environments are interleaved across teachers, frozen teacher policies provide target hand actions, and a single variational latent controller is regularized toward the tactile-conditioned prior. Held-out sphere scales (1.1 and 1.2) and additional object rows are not used as teachers.

Evaluation protocol. Unless otherwise stated, each reported row is evaluated over 1000 attempts with randomized evaluation seeds. Reorientation methods share the same target-orientation distribution and timeout cap within each benchmark. Axis-conditioned rotation uses a 20-second window and reports signed accumulated rotation projected onto the commanded axis; trials terminate early at object drop. Training curves report both episode reward and episode length to distinguish rotation progress from short-lived contact.

Sim-to-real and tactile interface. Simulation and hardware use the same deployable hand observation structure. In simulation, contact sensors on five elastomer links provide net contact forces and contact positions. Contact magnitude is the smoothed norm of recent contact-force history with a latency-style holdout model, and each 3D contact position is transformed into the corresponding tactile frame; inactive contacts are zeroed. On hardware, the tactile SDK returns per-finger 6D force and raw contact points. We use the norm of the first three force components as contact magnitude, threshold small forces to zero, select the highest-response contact point, and map its tactile image coordinate to a calibrated 3D fingertip point through tactile UV-to-3D maps. Real axis-conditioned deployment uses hand proprioception, tactile features, and command code, without object-pose camera feedback in the action loop.

Hardware transfer uses joint-level dynamics calibration followed by domain randomization. The real hand is excited with 1 Hz sinusoidal commands, and simulated K_p/K_d gains are adjusted to match the amplitude and phase of measured joint trajectories. The calibrated nominal model is combined with randomization over object mass, friction, PD gains, observation/action noise, reset pose, external perturbations, and gravity direction.

Downstream action interface. Downstream policies using TacBPM do not output low-level joint actions. They observe task information together with proprioceptive/tactile features, predict a residual latent code, add it to the frozen prior mean, and decode the resulting latent action into the 22-DoF hand command. In the strict-reuse setting both the prior and decoder are frozen. In the decoder-finetuning setting used for the main downstream transfer and axis-conditioned experiments, the prior remains fixed while the decoder and output head adapt to the downstream object geometry.

RL objectives. The repose objective measures target-pose correction under sustained contact. Its reward combines dense rotational progress, a goal-tolerance bonus, and penalties on

object-goal position drift, large actions, non-nominal hand posture, torque, mechanical work, and falls. The axis-conditioned objective replaces the target quaternion with a command vector. Rotation commands reward signed angular velocity along the commanded axis and penalize orthogonal angular velocity; the *still* command rewards holding the current pose and suppressing object motion. These objectives test finite-horizon goal correction and continuous command following through the same latent interface.

Baselines. The baselines isolate the action interface, tactile conditioning, and decoder adaptation. *RL from scratch* learns directly in the environment action space. *w/o tactile* keeps the residual-latent prior-decoder pipeline but zeros tactile force and contact-position channels before normalization. For complex-object transfer, we also compare actor-level transfer, action-space residual learning, a single-scale prior, and two TacBPM variants: frozen decoder for strict reuse and finetuned decoder with the prior fixed.

Metrics and ranking. Following prior dexterous reorientation and in-hand rotation work, we report success, drop/fallen rate, episode length, and rotation progress [11], [12], [14], [18]. Let q_t and q^g denote the object and goal quaternions, and let p_t and p^g denote the object and goal positions. The orientation error is the quaternion geodesic distance

$$e_R(t) = 2 \sin^{-1} \left(\min \left(1, \left\| \text{vec} \left(q_t \otimes (q^g)^{-1} \right) \right\|_2 \right) \right). \quad (8)$$

A reorientation trial is successful if $\min_t e_R(t) \leq 0.2$ rad before timeout or object drop. Object drop is defined by $e_p(t) = \|p_t - p^g\|_2 \geq 0.03$ m or by a height-based reset. Steps[†] counts failed attempts at the episode cap, so lower values reward both faster success and fewer failures. All table entries use mean $\pm$ 95% confidence intervals unless otherwise stated; highlighting marks the first, second, and third ranks by the mean under each metric direction.

Figure 2 summarizes the transition from multi-scale sphere distillation to downstream reuse on non-spherical objects.

B. Generalist Prior Coverage and Held-Out Extrapolation

We first test whether the distilled prior generalizes beyond its teacher scales. Figure 2(c) visualizes the latent space induced by online multi-teacher distillation, and Table II evaluates seen sphere scales, unseen sphere scales, and held-out object shapes. This experiment establishes whether the prior retains the teacher family before being used for downstream transfer.

Latent-space evidence. Table II provides the main causal evidence. Raw-action RL succeeds on the easiest mid-size sphere but is unstable across most other scales, indicating that direct exploration in the 22-D action space is a bottleneck. The learned prior-decoder interface without tactile input preserves much of the teacher family on seen scales, increasing average success from 26.03% to 93.67%. Tactile conditioning adds only a small gain on seen scales (93.67% to 95.62%), but becomes more important under contact-regime shift: on unseen scales and held-out shapes, TacBPM increases average success from 40.05% to 60.38% and reduces capped steps from 81.16 to 72.71. These results support two conclusions: distillation

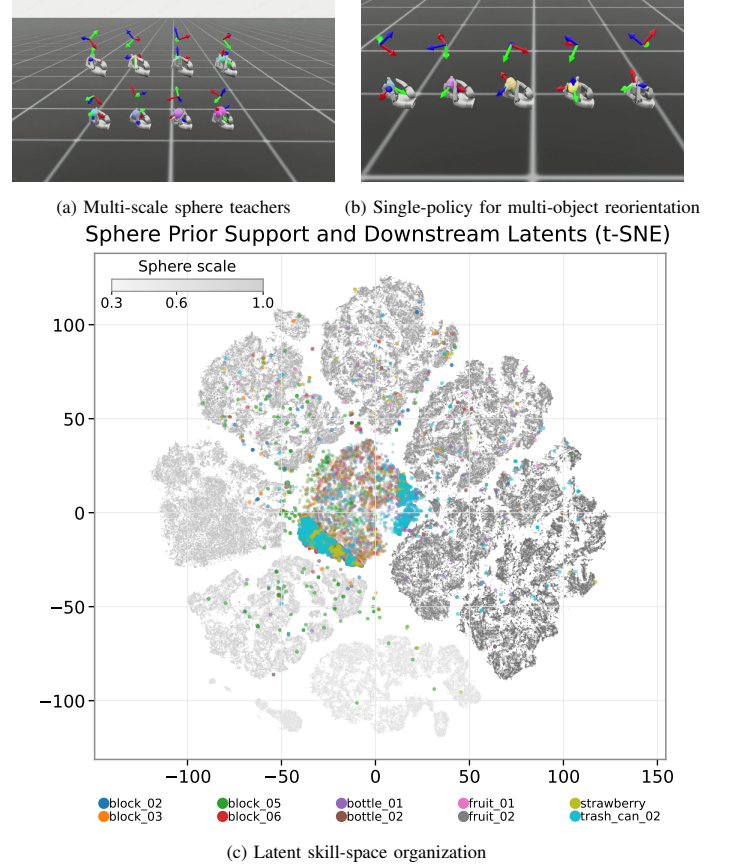


Fig. 2. Prior coverage and downstream reuse. The teacher rollouts in (a) expose the hand to sphere-scale variation used for distillation. The downstream rollout in (b) illustrates reuse of one prior-centered latent interface across object instances. The t-SNE visualization in (c) shows sphere-prior latent codes from all eight training scales and downstream executed residual latents for complex objects.

supplies a reusable latent action interface, and tactile contact features improve adaptation when geometry changes. The remaining gap on sphere (1.2) and several held-out shapes indicates the limit of extrapolating from a frozen sphere-trained prior.

C. Residual Latent Transfer to Complex Reorientation

We next evaluate downstream residual policies on anisotropic objects whose contact geometry differs from the sphere teachers. This setting tests whether the prior provides an effective action space for new object-specific policies rather than only reproducing sphere behavior. We compare strict reuse, where the decoder is frozen, with adaptive reuse, where the decoder is finetuned while the prior remains fixed. Table III reports raw-action PPO, actor-level transfer, action-space residual learning, single-scale priors, and TacBPM variants under the same reward, environment count, training budget, initialization, and evaluation protocol. Figure 4 shows the corresponding learning curves.

Table III separates action-space structure, prior diversity, and decoder adaptation. Raw-action PPO rarely succeeds (0.80–2.10% on single objects and 1.00% in Multi), showing the difficulty of low-level exploration. Actor-level transfer is

TABLE II
SPHERE-SCALE AND UNSEEN-OBJECT EVALUATION OF THE GENERALIST PRIOR ABLATIONS. SUCCESS RATES ARE REPORTED WITH 95% CONFIDENCE INTERVALS. STEPS[†] COUNTS FAILED ATTEMPTS AS 100 STEPS. CASES SPHERE (1.1) AND SPHERE (1.2) ARE UNSEEN SPHERE SCALES, AND THE ADDITIONAL OBJECT ROWS EVALUATE HELD-OUT OBJECT SHAPES OUTSIDE THE TEACHER SET.

Case	From scratch		w/o tactile		Ours	
	Succ. $\uparrow$	Steps [†] $\downarrow$	Succ. $\uparrow$	Steps [†] $\downarrow$	Succ. $\uparrow$	Steps [†] $\downarrow$
sphere (0.3)	24.50 $\pm$ 2.67%	79.40 $\pm$ 2.27	92.90 $\pm$ 1.59%	35.03 $\pm$ 1.59	97.30$\pm$1.01%	31.07$\pm$1.26
sphere (0.4)	25.20 $\pm$ 2.69%	79.81 $\pm$ 2.21	96.10 $\pm$ 1.20%	29.76 $\pm$ 1.16	98.40$\pm$0.78%	27.58$\pm$0.94
sphere (0.5)	99.80 $\pm$ 0.28%	3.28$\pm$0.30	100.00$\pm$0.00%	3.42 $\pm$ 0.18	99.90 $\pm$ 0.20%	3.45 $\pm$ 0.27
sphere (0.6)	20.40 $\pm$ 2.50%	84.18 $\pm$ 2.00	96.50 $\pm$ 1.14%	33.33 $\pm$ 1.26	98.90$\pm$0.65%	29.72$\pm$0.98
sphere (0.7)	14.60 $\pm$ 2.19%	89.27 $\pm$ 1.67	94.20 $\pm$ 1.45%	38.31 $\pm$ 1.43	97.00$\pm$1.06%	35.29$\pm$1.11
sphere (0.8)	10.20 $\pm$ 1.88%	92.95 $\pm$ 1.47	92.90 $\pm$ 1.59%	41.73 $\pm$ 1.36	96.80$\pm$1.09%	39.34$\pm$1.17
sphere (0.9)	8.40 $\pm$ 1.72%	93.95 $\pm$ 1.30	91.40 $\pm$ 1.74%	50.63 $\pm$ 1.50	92.00$\pm$1.68%	49.81$\pm$1.50
sphere (1.0)	5.10 $\pm$ 1.36%	96.11 $\pm$ 1.05	85.40$\pm$2.19%	61.13 $\pm$ 1.66	84.70 $\pm$ 2.23%	60.88$\pm$1.66
Seen Avg.	26.03 $\pm$ 21.29%	77.37 $\pm$ 21.21	93.67 $\pm$ 2.97%	36.67 $\pm$ 11.68	95.62$\pm$3.47%	34.64$\pm$11.72
sphere (1.1)	4.90 $\pm$ 1.34%	96.20 $\pm$ 1.05	55.50 $\pm$ 3.08%	85.10 $\pm$ 1.39	62.00$\pm$3.01%	83.10$\pm$1.49
sphere (1.2)	3.30 $\pm$ 1.11%	97.52$\pm$0.84	1.70 $\pm$ 0.80%	100.04 $\pm$ 0.44	13.30$\pm$2.11%	99.80 $\pm$ 0.98
block_01	18.30 $\pm$ 2.40%	85.26 $\pm$ 1.95	58.20 $\pm$ 3.06%	72.56 $\pm$ 2.07	79.50$\pm$2.50%	61.10$\pm$1.92
trash_can_01	11.70 $\pm$ 1.99%	90.62 $\pm$ 1.61	14.60 $\pm$ 2.19%	94.56 $\pm$ 1.25	79.50$\pm$0.00%	61.10$\pm$0.00
switch_01	13.80 $\pm$ 2.14%	89.30 $\pm$ 1.68	28.70 $\pm$ 2.81%	85.27 $\pm$ 1.72	47.10$\pm$3.10%	80.49$\pm$1.90
box_01	13.50 $\pm$ 2.12%	89.69 $\pm$ 1.65	29.00 $\pm$ 2.81%	90.24 $\pm$ 1.68	41.80$\pm$3.06%	88.24$\pm$1.99
bottle_03	11.80 $\pm$ 2.00%	90.55 $\pm$ 1.62	57.70 $\pm$ 3.06%	69.21 $\pm$ 1.91	77.80$\pm$2.58%	59.99$\pm$1.79
strawberry	14.50 $\pm$ 2.18%	88.40 $\pm$ 1.76	75.00 $\pm$ 2.69%	52.26 $\pm$ 1.87	82.00$\pm$2.38%	47.83$\pm$1.69
Unseen Avg.	11.47 $\pm$ 3.47%	90.94 $\pm$ 2.80	40.05 $\pm$ 17.51%	81.16 $\pm$ 10.81	60.38$\pm$17.06%	72.71$\pm$12.27

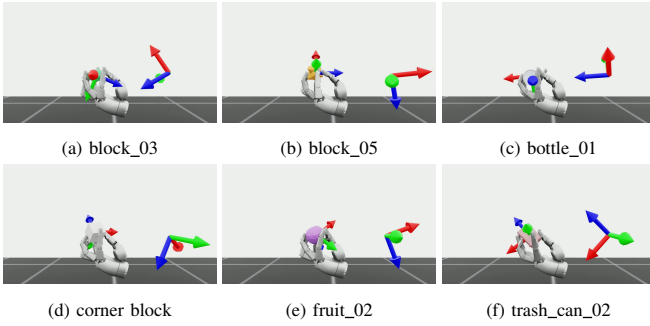


Fig. 3. Qualitative examples from downstream reorientation on complex objects. The six objects differ in size, shape, and contact geometry, but the downstream controller uses the same prior-centered residual latent interface.

stronger but transfers an executable specialist actor; its Multi success is 40.00%, below both TacBPM variants. Action-space residual learning reaches 82.40% on fruit_02 but drops to 8.80% in Multi, suggesting that low-level residuals can fit favorable single geometries but do not preserve contact strategies across object changes. The single-scale prior is a strong baseline, indicating that prior distillation itself is important; however, its Multi success (65.80%) remains below the multi-scale TacBPM variants (73.30% frozen decoder and 70.00% finetuned decoder), supporting the value of scale-diverse teacher coverage.

The frozen and finetuned decoder rows evaluate strict reuse versus geometry adaptation. Decoder finetuning gives the best single-object success on block_03, block_05, bottle_01, and trash_can_02, improving over the frozen decoder by 5.8, 6.7, 2.7, and 1.8 percentage points. This indicates that the fixed prior can be retained while the decoder adapts latent commands to local geometry. Conversely, the frozen decoder achieves the best Multi success (73.30%), suggesting that strict decoder reuse regularizes a shared policy over broader object variation. The reward and episode-length curves in Fig. 4 further show that TacBPM variants improve while maintaining longer controlled interactions.

D. Axis-Conditioned Rotation

Axis-conditioned rotation tests whether the same latent interface can be directed by task commands. Unlike goal-conditioned reorientation, the policy must rotate the object about a signed Cartesian axis rather than merely reduce orientation error. The downstream policy receives a repeated 7-way command code for $+x$, $-x$, $+y$, $-y$, $+z$, $-z$, and *still*. Commands are periodically resampled during training, requiring one policy to switch between incompatible rotation intents while maintaining contact. Rotate commands reward signed angular velocity along the commanded axis and penalize off-axis rotation; the *still* command rewards pose holding and motion suppression. The prior remains fixed and the decoder follows the downstream finetuning setup. Table IV reports the macro-average signed commanded-axis angle over the six rotation commands.

Across the six simulated objects with no-tactile evaluations, TacBPM obtains the highest axis-average rotation. The gains are clearest on block_08, block_14, and bottle_02, where from-scratch learning nearly collapses but residual-latent policies continue to produce commanded-axis motion. The no-tactile variant often exceeds raw-action exploration, confirming that the prior-decoder interface contributes independently of tactile input. Its consistent gap to TacBPM indicates that tactile conditioning contributes additional contact-state information, especially when geometry and support modes vary. Rows with large confidence intervals, such as trash_can_02, should be interpreted as difficult cases with occasional high-progress rollouts rather than uniformly stable rotation; future reporting should include separate off-axis and drop-rate metrics.

The real-robot results in Fig. 6 and Table V show a similar trend under hardware contact noise. TacBPM performs best on most object-axis pairs and is strongest overall on corner block, standard tennis, and the unseen multiface object. The small-tennis exceptions favor the no-tactile policy in accumulated angle, but qualitative rollouts show less stable contact, including slip, drops, and stuck configurations. From-scratch policies exhibit a different failure pattern: they can rotate along some axes but collapse on others, suggesting axis-specific local optima from raw-action exploration. Matched videos for the unseen multiface object further show that TacBPM maintains rolling contact more consistently than the baselines. These videos are provided on the project website¹.

E. Arm-Hand Embodiment Transfer

Arm-hand transfer evaluates the hand prior when dexterous control is coupled to arm motion. In this setting, the arm brings the object into a reachable interaction regime, while the hand uses the same residual latent interface as in the in-hand experiments. This changes approach direction, gravity loading, and contact availability. We include this as a planned embodiment-transfer benchmark comparing raw arm-hand PPO, a modular latent-decoder baseline, and arm control coupled with TacBPM. The expected limitation is that the hand

¹Project videos: <https://tacbpm.github.io>.

TABLE III
DOWNSTREAM REORIENTATION ON COMPLEX OBJECTS. THE PRIOR IS FIXED FOR TACBPM; DOWNSTREAM VARIANTS EITHER FREEZE OR FINETUNE THE DECODER WHILE TRAINING THE RESIDUAL LATENT POLICY. MULTI JOINTLY EVALUATES BLOCK_02, BLOCK_06, FRUIT_01, BOTTLE_02, AND SPHERE_02.

Method	block_03		block_05		bottle_01		fruit_02		trash_can_02		Multi	
	Succ. $\uparrow$	Steps $\downarrow$	Succ. $\uparrow$	Steps $\downarrow$	Succ. $\uparrow$	Steps $\downarrow$	Succ. $\uparrow$	Steps $\downarrow$	Succ. $\uparrow$	Steps $\downarrow$	Succ. $\uparrow$	Steps $\downarrow$
RL from scratch	1.40 $\pm$ 0.73%	98.73 $\pm$ 0.66	2.10 $\pm$ 0.89%	98.09 $\pm$ 0.81	1.70 $\pm$ 0.80%	98.42 $\pm$ 0.74	0.80 $\pm$ 0.55%	99.26 $\pm$ 0.52	0.90 $\pm$ 0.59%	99.17 $\pm$ 0.54	1.00 $\pm$ 0.62%	99.08 $\pm$ 0.57
RL with pretrained actor	70.40 $\pm$ 2.83%	42.43 $\pm$ 2.34	54.60 $\pm$ 3.09%	54.33 $\pm$ 2.59	60.70 $\pm$ 3.03%	50.97 $\pm$ 2.47	73.00 $\pm$ 2.75%	43.13 $\pm$ 2.16	44.10 $\pm$ 3.08%	63.77 $\pm$ 2.55	40.00 $\pm$ 3.04%	69.76 $\pm$ 2.33
Action-space residual	73.20 $\pm$ 2.75%	42.98 $\pm$ 2.17	52.20 $\pm$ 3.10%	57.37 $\pm$ 2.54	60.40 $\pm$ 3.03%	54.03 $\pm$ 2.34	82.40$\pm$2.36%	36.41$\pm$1.85	47.80 $\pm$ 3.10%	64.67 $\pm$ 2.33	8.80 $\pm$ 1.76%	93.54 $\pm$ 1.30
Single-scale prior	73.70 $\pm$ 2.73%	41.10 $\pm$ 2.21	59.50 $\pm$ 3.04%	50.02 $\pm$ 2.57	65.20 $\pm$ 2.95%	48.01 $\pm$ 2.38	72.20 $\pm$ 2.78%	44.16 $\pm$ 2.17	67.20 $\pm$ 2.91%	46.45 $\pm$ 2.35	65.80 $\pm$ 2.94%	44.27 $\pm$ 2.51
TacBPM w. frozen decoder	72.80 $\pm$ 2.70%	40.65 $\pm$ 2.27	61.40 $\pm$ 3.02%	48.64 $\pm$ 2.54	67.40 $\pm$ 2.91%	46.57 $\pm$ 2.33	72.70 $\pm$ 2.76%	43.99 $\pm$ 2.15	72.00 $\pm$ 2.78%	43.75 $\pm$ 2.20	73.30$\pm$2.74%	41.69 $\pm$ 2.20
TacBPM w. finetuning decoder	78.60$\pm$2.54%	35.76$\pm$2.11	68.10$\pm$2.89%	42.47$\pm$2.45	70.10$\pm$2.84%	43.77$\pm$2.30	75.20 $\pm$ 2.68%	41.32 $\pm$ 2.11	73.80$\pm$2.73%	43.19$\pm$2.14	70.00 $\pm$ 2.84%	39.23$\pm$2.48

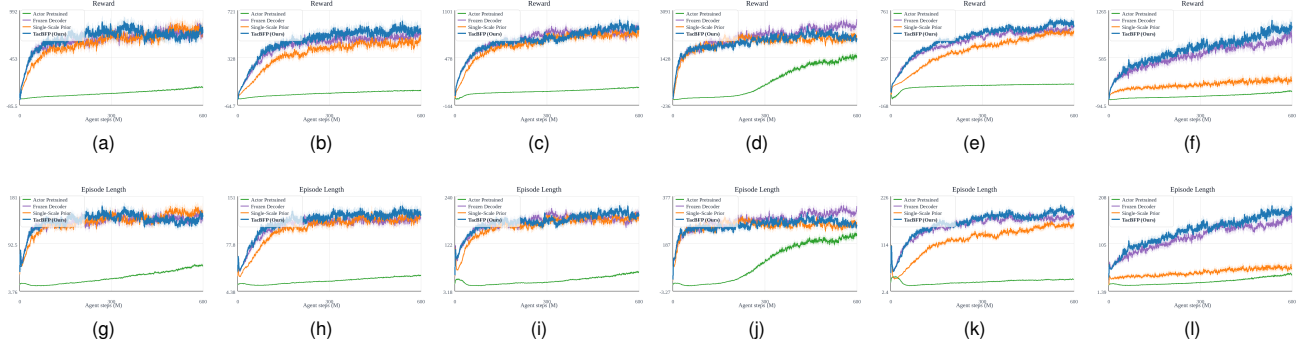


Fig. 4. Downstream reposition training curves for the complex-object transfer ablations. The columns, from left to right, correspond to block_03, block_05, bottle_01, fruit_02, trash_can_02, and Multi. Subfigures (a)–(f) show episode reward, and (g)–(l) show episode length. Curves are plotted up to 600M agent steps; shaded regions indicate rolling 95% confidence intervals.

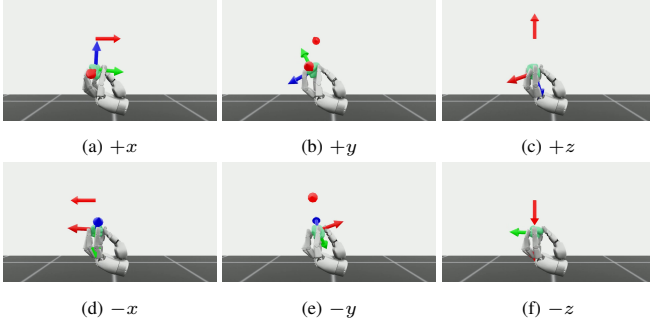


Fig. 5. Axis-conditioned simulation examples. A single residual-latent policy receives signed Cartesian rotation commands and must produce object motion along the requested axis while maintaining stable in-hand contact.

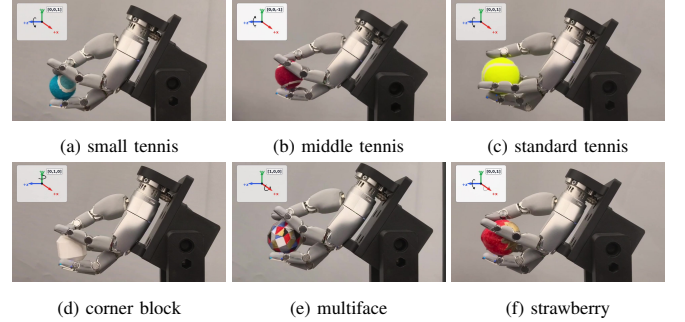


Fig. 6. Real-robot axis-conditioned rollouts. The same evaluation protocol is applied to physical objects with different size, geometry, and compliance, testing whether the learned tactile behavior prior transfers outside simulation.

TABLE IV
AXIS-CONDITIONED DOWNSTREAM ROTATION ON THE SIX OBJECTS WITH NO-TACTILE EVALUATIONS. AXIS AVG. IS THE MACRO-AVERAGE SIGNED COMMANDED-AXIS ANGLE OVER THE SIX COMMANDED AXES, REPORTED IN DEGREES AS MEAN $\pm$ 95% CONFIDENCE INTERVAL. HIGHER IS BETTER.

Object	From scratch	w/o tactile	Ours
trashcan_02	76.63 $\pm$ 42.48	48.74 $\pm$ 16.78	225.61$\pm$169.78
strawberry	59.44 $\pm$ 14.64	44.10 $\pm$ 13.64	69.39$\pm$10.56
bottle_02	−0.43 $\pm$ 14.65	181.68 $\pm$ 56.49	192.98$\pm$43.85
block_03	64.34 $\pm$ 12.42	75.24 $\pm$ 18.87	78.28$\pm$25.67
block_08	1.29 $\pm$ 18.48	80.21 $\pm$ 11.50	117.39$\pm$27.80
block_14	0.00 $\pm$ 9.94	49.55 $\pm$ 21.90	77.06$\pm$31.96

prior can stabilize contact after object presentation, but large approach errors remain the responsibility of the arm policy.

F. Online Reinforcement Learning and Imitation Learning Interfaces

The final planned evaluation studies TacBPM as a learning interface beyond massively parallel downstream PPO. The fixed prior and decoder separate reusable hand behavior from task-specific decision making: the prior defines plausible contact-rich actions for the current tactile-proprioceptive state, while the high-level learner chooses the residual intent. This separation is particularly relevant for single-environment online RL, where random low-level hand actions often lose contact before reward is observed. The same interface can also support imitation learning by mapping demonstration- or perception-conditioned residual latents to contact-aware hand actions.

Future low-data settings will also clarify several design choices. Auxiliary object-state prediction may encourage the high-level policy to encode object motion without changing

TABLE V

REAL-ROBOT AXIS-CONDITIONED ROTATION. EACH ENTRY REPORTS THE MEAN SIGNED COMMANDED-AXIS ROTATION ANGLE OVER 10 EPISODES. EACH EPISODE RUNS FOR UP TO 20 SECONDS AND TERMINATES EARLY IF THE OBJECT IS DROPPED. HIGHER IS BETTER.

Object	Method	+x ↑	-x ↑	+y ↑	-y ↑	+z ↑	-z ↑
corner block	From scratch	428°	162°	27°	81°	293°	113°
	w/o tactile	59°	22°	31°	13°	257°	30°
	Ours	644°	524°	750°	441°	806°	405°
small tennis	From scratch	243°	171°	54°	149°	441°	342°
	w/o tactile	337°	528°	19°	14°	576°	473°
	Ours	765°	519°	383°	338°	945°	394°
standard tennis	From scratch	70°	40°	288°	38°	216°	342°
	w/o tactile	59°	41°	202°	132°	218°	239°
	Ours	378°	192°	495°	137°	456°	633°
multiface object (unseen)	From scratch	158°	22°	315°	279°	167°	16°
	w/o tactile	258°	297°	23°	48°	162°	239°
	Ours	675°	459°	797°	324°	754°	799°

TABLE VI

ARM-HAND MANIPULATION TRANSFER. THE TABLE MEASURES WHETHER THE FROZEN HAND PRIOR REMAINS EFFECTIVE WHEN MANIPULATION ALSO REQUIRES ARM MOTION.

Method	Task Success ↑	Drop Rate ↓	Steps ↓	Reward ↑
Raw arm-hand PPO				
Arm + hand latent decoder				
Arm + TacBPM				

the deployed observation stream. Residual regularization can control deviations from the prior mean. Richer temporal models, multimodal latent distributions, and tactile dropout can further test robustness under sensor degradation.

VI. CONCLUSION

We presented TacBPM, a tactile-conditioned behavior prior model for dexterous in-hand reorientation. TacBPM distills eight sphere-scale specialists into a variational latent controller with a tactile-proprioceptive prior and decoder. Downstream policies act through residual latent codes around the fixed prior mean, while the decoder is either frozen for strict reuse or finetuned for geometry adaptation. Results on held-out scales, anisotropic objects, commanded-axis rotation, and real-robot rollouts support the use of sphere-trained contact behaviors as a reusable prior-centered action interface for reorientation-style tasks. Future work will evaluate this interface in arm-hand manipulation, single-environment online RL, and imitation learning.

REFERENCES

- [1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [2] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, “Isaac gym: High performance gpu-based physics simulation for robot learning,” *arXiv preprint arXiv:2108.10470*, 2021.
- [3] Z. Luo, J. Cao, J. Merel, A. Winkler, J. Huang, K. Kitani, and W. Xu, “Universal humanoid motion representations for physics-based control,” in *International Conference on Learning Representations*, 2024.
- [4] X. B. Peng, Y. Guo, L. Halper, S. Levine, and S. Fidler, “ASE: Large-scale reusable adversarial skill embeddings for physically simulated characters,” *arXiv preprint arXiv:2205.01906*, 2022.

TABLE VII

PLANNED EVALUATIONS OF TACBPM AS A LEARNING INTERFACE. THE TACTILE PRIOR REMAINS FIXED WHILE THE HIGH-LEVEL LEARNER, DECODER ADAPTATION, AND DATA-COLLECTION REGIME CHANGE.

Setting	Baseline	Evaluation question
Single-env online RL Imitation learning	Raw action PPO Direct action BC	Does the latent interface reduce exploration burden? Does the decoder improve contact stability from demonstrations?

- [5] C. Tessler, Y. Kasten, Y. Guo, S. Mannor, G. Chechik, and X. B. Peng, “CALM: Conditional adversarial latent models for directable virtual characters,” in *ACM SIGGRAPH Conference Proceedings*, 2023.
- [6] J. Merel, L. Hasenclever, A. Galashov, A. Ahuja, V. Pham, G. Wayne, Y. W. Teh, and N. Heess, “Neural probabilistic motor primitives for humanoid control,” *arXiv preprint arXiv:1811.11711*, 2018.
- [7] H. Yao, Z. Song, B. Chen, and L. Liu, “ControlVAE: Model-based learning of generative controllers for physics-based characters,” *arXiv preprint arXiv:2210.06063*, 2022.
- [8] J. Won, D. Gopinath, and J. Hodgins, “Physics-based character controllers using conditional VAEs,” *ACM Transactions on Graphics*, vol. 41, no. 4, pp. 1–12, 2022.
- [9] Q. Zhu, H. Zhang, M. Lan, and L. Han, “Neural categorical priors for physics-based character control,” *arXiv preprint arXiv:2308.07200*, 2023.
- [10] A. Rajeswaran, V. Kumar, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, and S. Levine, “Learning complex dexterous manipulation with deep reinforcement learning and demonstrations,” in *Proceedings of Robotics: Science and Systems*, 2018.
- [11] OpenAI, M. Andrychowicz, B. Baker, M. Chociej, R. Jozefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray, J. Schneider, S. Sidor, J. Tobin, P. Welinder, L. Weng, and W. Zaremba, “Learning dexterous in-hand manipulation,” *The International Journal of Robotics Research*, vol. 39, no. 1, pp. 3–20, 2020.
- [12] T. Chen, J. Xu, and P. Agrawal, “A system for general in-hand object re-orientation,” in *Proceedings of the 5th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 164, 2022, pp. 297–307. [Online]. Available: <https://proceedings.mlr.press/v164/chen22a.html>
- [13] Z.-H. Yin, B. Huang, Y. Qin, Q. Chen, and X. Wang, “Rotating without seeing: Towards in-hand dexterity through touch,” in *Proceedings of Robotics: Science and Systems*, 2023.
- [14] H. Qi, B. Yi, S. Suresh, M. Lambeta, Y. Ma, R. Calandra, and J. Malik, “General in-hand object rotation with vision and touch,” in *Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 229, 2023, pp. 2549–2564. [Online]. Available: <https://mlanthology.org/corl/2023/qi2023corl-general/>
- [15] Y. Yuan, H. Che, Y. Qin, B. Huang, Z.-H. Yin, K.-W. Lee, Y. Wu, S.-C. Lim, and X. Wang, “Robot synesthesia: In-hand manipulation with visuotactile sensing,” in *IEEE International Conference on Robotics and Automation*, 2024, pp. 6558–6565.
- [16] J. Yin, H. Qi, J. Malik, J. Pikul, M. Yim, and T. Hellebrekers, “Learning in-hand translation using tactile skin with shear and normal force sensing,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 5850–5856.
- [17] A. Handa, A. Allshire, V. Makoviychuk, A. Petrenko, R. Singh, J. Liu, D. Makoviychuk, K. Van Wyk, A. Zhurkevich, B. Sundaralingam, Y. S. Narang, J.-F. Lafleche, D. Fox, and G. State, “Dextreme: Transfer of agile in-hand manipulation from simulation to reality,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 5977–5984.
- [18] H. Qi, A. Kumar, R. Calandra, Y. Ma, and J. Malik, “In-hand object rotation via rapid motor adaptation,” in *Conference on Robot Learning*, 2022.
- [19] A. Bhardwaj, M. Wilder-Smith, M. Mittal, V. Patil, and M. Hutter, “Viserdex: Visual sim-to-real for robust dexterous in-hand reorientation,” 2026.
- [20] M. Yang, C. Lu, A. Church, Y. Lin, C. Ford, H. Li, E. Psomopoulou, D. A. W. Barton, and N. F. Lepora, “Anyrotate: Gravity-invariant in-hand object rotation with sim-to-real touch,” *arXiv preprint arXiv:2405.07391*, 2024.
- [21] X. Liu, H. Wang, and L. Yi, “Dexndm: Closing the reality gap for dexterous in-hand rotation via joint-wise neural dynamics model,” *arXiv preprint arXiv:2510.08556*, 2025.
- [22] J. Wang, Y. Yuan, H. Che, H. Qi, Y. Ma, J. Malik, and X. Wang, “Lessons from learning to spin pens,” *arXiv preprint arXiv:2407.18902*, 2024.

- [23] I. Guzey, B. Evans, S. Chintala, and L. Pinto, “Dexterity from touch: Self-supervised pre-training of tactile representations with robotic play,” in *Proceedings of the 7th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 229, 2023, pp. 3142–3166. [Online]. Available: <https://proceedings.mlr.press/v229/guzey23a.html>
- [24] S. Suresh, H. Qi, T. Wu, T. Fan, L. Pineda, M. Lambeta, J. Malik, M. Kalakrishnan, R. Calandra, M. Kaess, J. Ortiz, and M. Mukadam, “Neuralfeels with neural fields: Visuotactile perception for in-hand manipulation,” *Science Robotics*, vol. 9, no. 96, p. eadl0628, 2024.
- [25] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, “3D-ViTac: Learning fine-grained manipulation with visuo-tactile sensing,” in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2024, pp. 2557–2578. [Online]. Available: <https://mlanthology.org/corl/2024/huang2024corl-3dvtac/>
- [26] A. A. Rusu, S. G. Colmenarejo, C. Gulcehre, G. Desjardins, J. Kirkpatrick, R. Pascanu, V. Mnih, K. Kavukcuoglu, and R. Hadsell, “Policy distillation,” in *International Conference on Learning Representations Workshop*, 2016.
- [27] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” *arXiv preprint arXiv:1011.0686*, 2010.
- [28] H. Qi, B. Yi, M. Lambeta, Y. Ma, R. Calandra, and J. Malik, “From simple to complex skills: The case of in-hand object reorientation,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 14 291–14 298.
- [29] J. Wan, X. Liu, and Y. Dong, “Dexremoe: In-hand reorientation of general object via mixtures of experts,” 2025.